To: Head of Product, Scout
From: Edward Ge
Subject: Product Prioritization for Recommendations and Price-Drop Alerts
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Business Question

Which products should Scout prioritize for personalized recommendations and price-drop alerts?

Recommendation

Scout should separate product prioritization into two tracks. For personalized recommendations, Scout should prioritize products that have both strong revenue and stable order volume, rather than only choosing the products with the highest total sales. For price-drop alerts, Scout should focus on higher-ticket products that still have enough order history to show real demand.

Based on the analysis, the strongest recommendation candidates include **WHITE HANGING HEART T-LIGHT HOLDER**, **JUMBO BAG RED RETROSPOT**, **PARTY BUNTING**, **ASSORTED COLOUR BIRD ORNAMENT**, and **REGENCY CAKESTAND 3 TIER**. These products performed well after ranking products by a weighted recommendation score using revenue, order count, and quantity sold.

For price-drop alerts, Scout should test higher-priced home and furniture-style products, such as **RUSTIC SEVENTEEN DRAWER SIDEBOARD**, **VINTAGE BLUE KITCHEN CABINET**, **VINTAGE RED KITCHEN CABINET**, **LOVE SEAT ANTIQUE WHITE METAL**, and **SET/4 WHITE RETRO STORAGE CUBES**. These products have higher average unit prices, so a discount or price change may be more meaningful to shoppers.

Evidence

I used the UCI Online Retail dataset, which contains transaction-level records from a UK-based online retailer. After removing cancelled invoices, invalid quantities, and non-positive prices, the cleaned dataset contained **530,104** transaction rows. I also removed non-product records such as postage, Amazon fees, bank charges, manual adjustments, and bad debt adjustments from the product-level analysis. These records accounted for only **0.41% of cleaned rows**, but about **3.63% of cleaned revenue**, so removing them helped avoid treating service or accounting items as recommendable products.

One issue with ranking only by total revenue is that it can over-prioritize one-off bulk purchases. For example, **PAPER CRAFT, LITTLE BIRDIE** generated very high revenue, but it appeared in only one order. This does not look like stable customer demand, so I did not treat revenue alone as enough evidence for recommendation priority.

To address this, I created a simple recommendation score based on revenue rank, order count rank, and quantity rank. This favored products that were not only valuable, but also repeatedly purchased. Under this approach, products like **WHITE HANGING HEART T-LIGHT HOLDER**, **JUMBO BAG RED RETROSPOT**, and **PARTY BUNTING** ranked highly because they combined strong sales with broader order activity.

For price-drop alerts, I used a different logic. Products with high average unit prices and at least 10 orders were treated as better candidates because customers may care more about price changes when the item is more expensive. This is why higher-ticket home goods and furniture-like products were selected for the price-alert track.

Limitations and Next Steps

This dataset does not include browsing behavior, saved items, wishlist activity, or actual price-change history. Because of that, this analysis should be treated as a product prioritization exercise rather than proof of recommendation performance. If Scout had internal data, I would next validate these products using user saves, clicks, alert opens, and conversions after price-drop notifications.